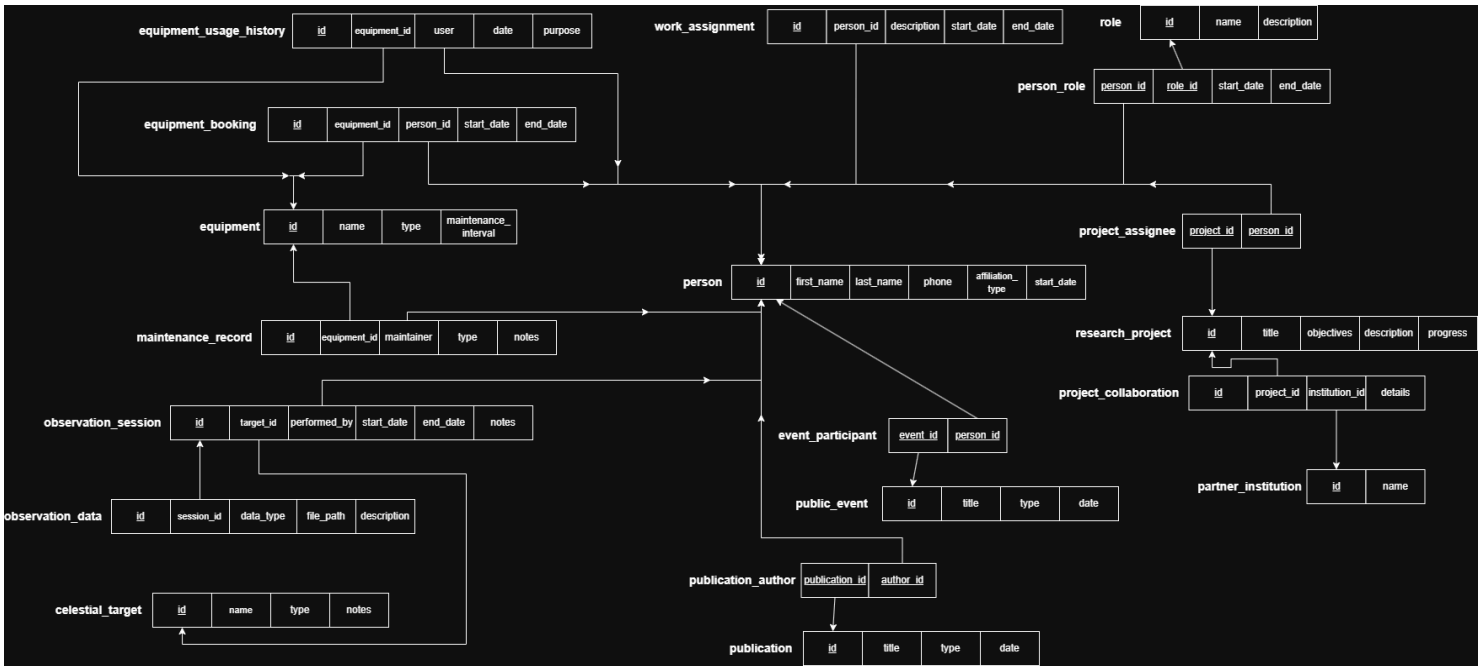
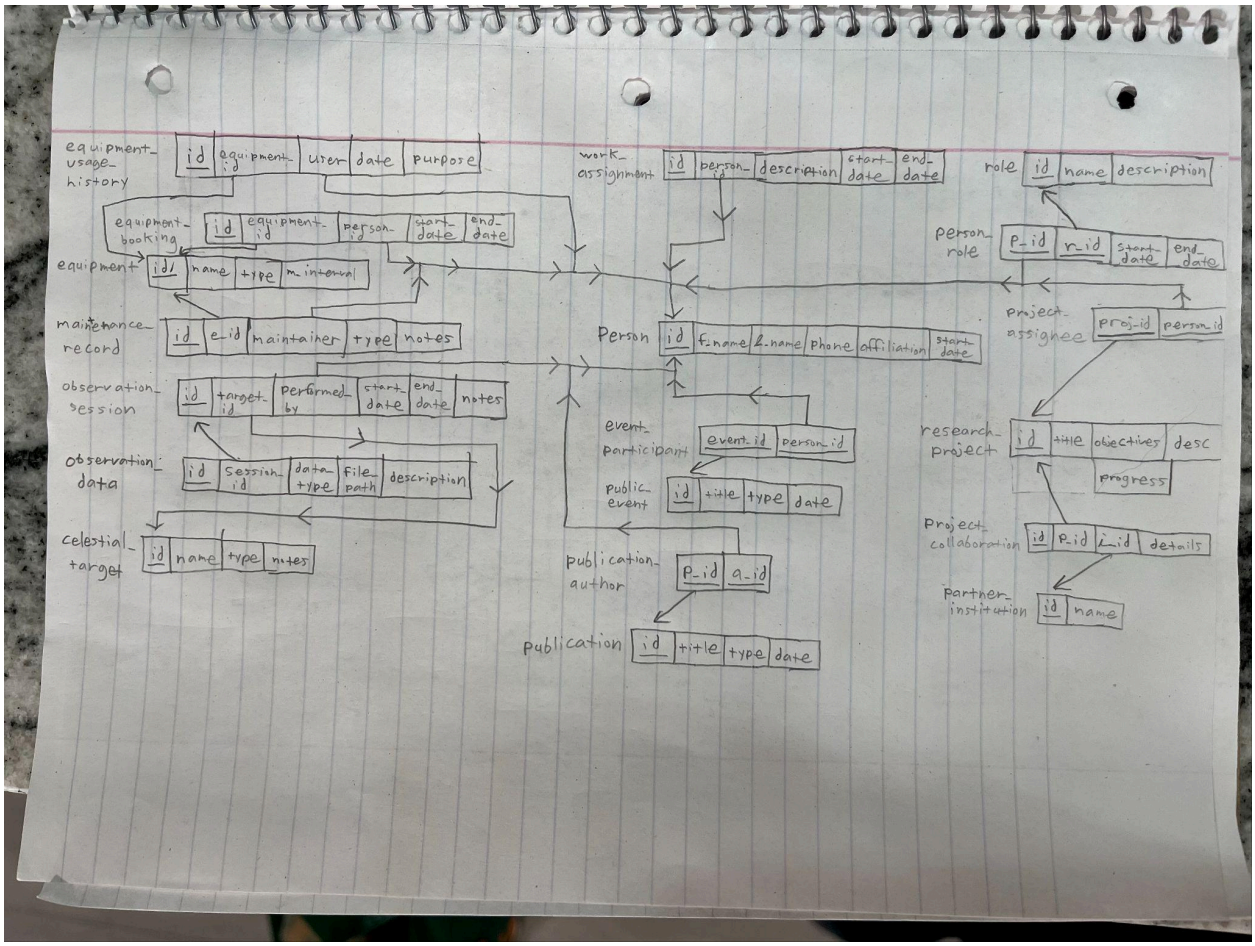
Tyler Millershaski
Schema Diagram

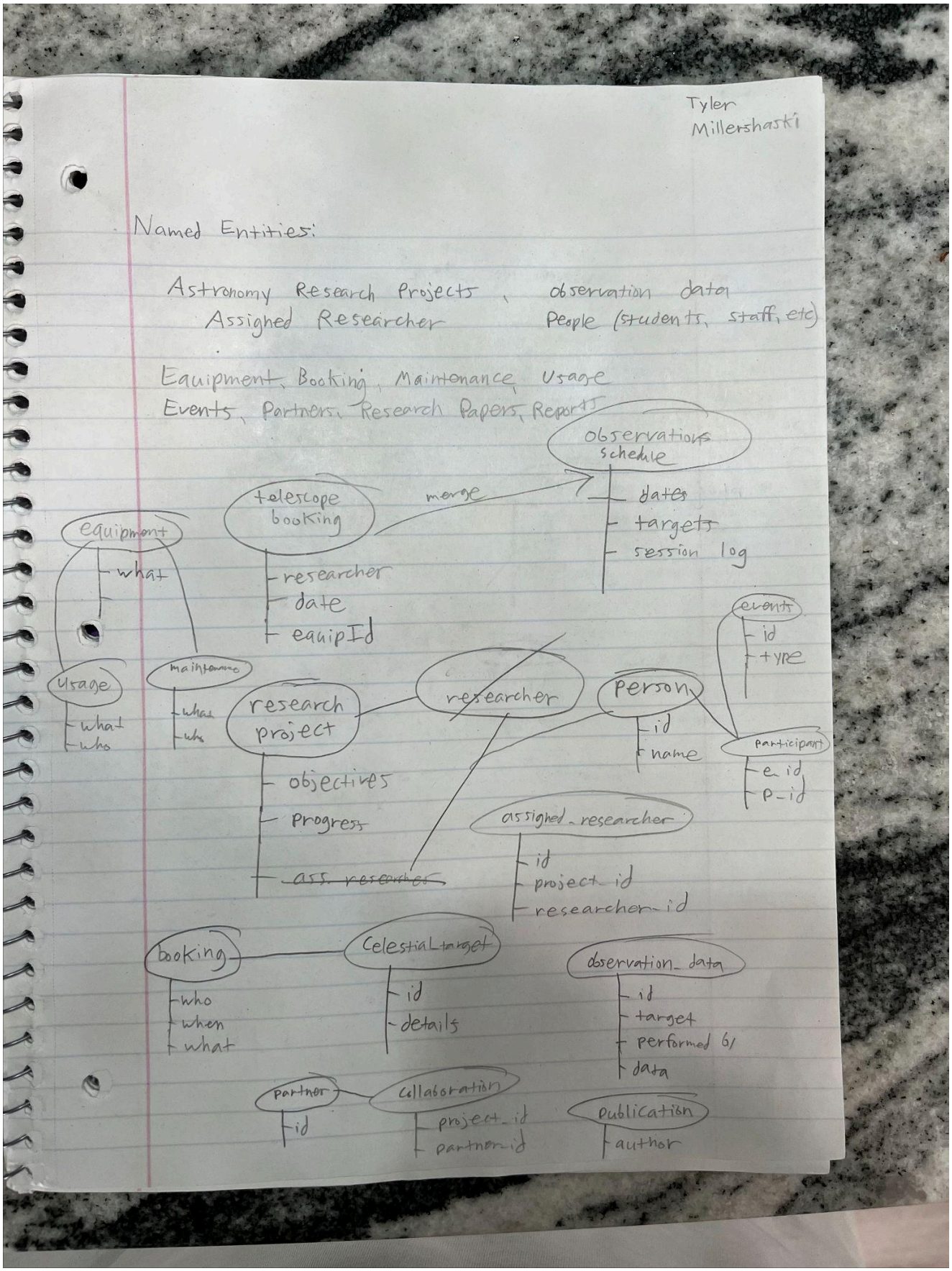
Final Schema (Draw.io)



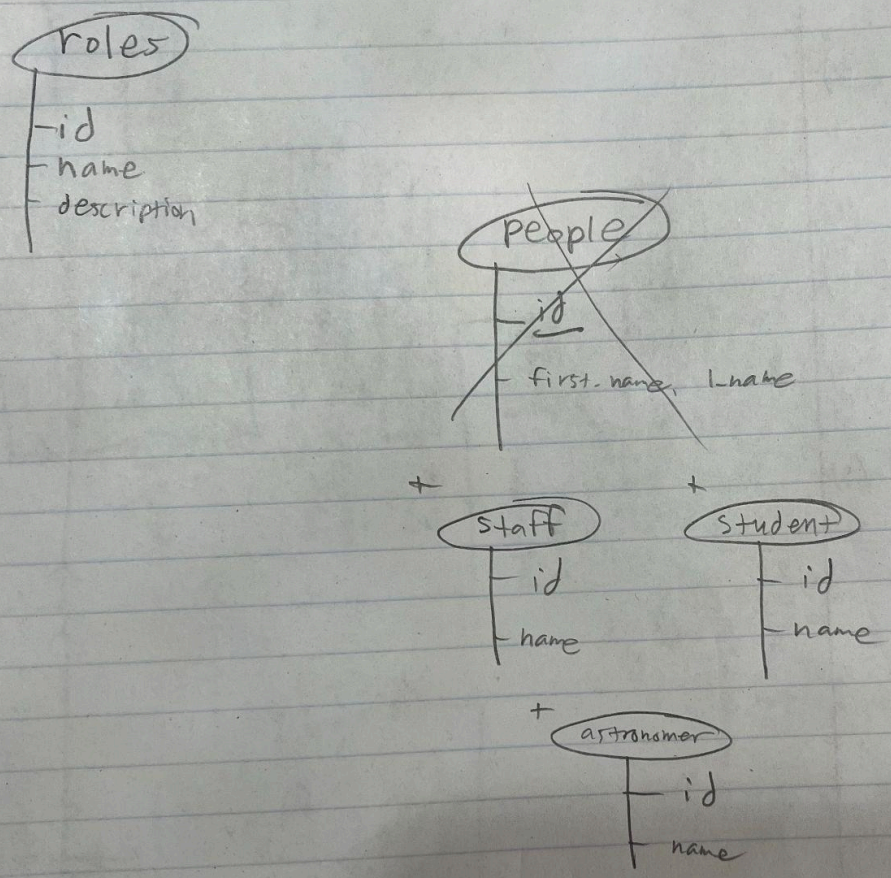
Paper Schema



Paper Work


Tyler
Millershaski
Named Entities:
Astronomy Research Projects , observation data
Assigned Researcher People (students, staff, etc)
Equipment, Booking, Maintenance, Usage
Events, Partners, Research Papers, Reports
observations schedule
dates
targets
session log
telescope booking
merge
researcher
date
equipId
equipment
what
Usage
what
who
maintenance
what
who
research project
objectives
Progress
~~ass. researcher~~
~~researcher~~
person
id
name
events
id
type
participant
e_id
p_id
assigned_researcher
id
project_id
researcher_id
booking
who
when
what
Celestial target
id
details
observation_data
id
target
performed by
data
Partner
id
Collaboration
project_id
partner_id
Publication
author

make all entities singular



Sample Data

role

id	name	description
1	Lead Astronomer	Oversees observation planning and research direction
2	Research Assistant	Supports data collection and analysis
3	Volunteer Coordinator	Manages volunteer participation for events

person

id	first_name	last_name	phone	affiliation_type	start_date
1	Amina	Patel	555-0101	Staff	2024-01-15
2	Leo	Martinez	555-0102	Researcher	2024-03-01
3	Sofia	Nguyen	555-0103	Volunteer	2024-06-20

person_role

person_id	role_id	start_date	end_date
1	1	2024-01-15	2024-03-01
2	2	2024-03-01	2024-03-01
3	3	2024-06-20	2024-06-20

work_assignment

id	person_id	description	start_date	end_date
1	1	Coordinate monthly deep-sky observation program	2025-01-01	2025-06-30
2	2	Analyze spectral data from nebula sessions	2025-02-15	2025-08-31
3	3	Support public viewing logistics and setup	2025-04-01	2025-10-31

research_project

id	title	objectives	description	progress
1	Exoplanet Transit Survey	Detect transit signatures	Monitoring candidate stars for periodic dimming	42.5
2	Nebula Spectral Atlas	Catalog spectral fingerprints	Building a reference atlas for nearby nebulae	68
3	Lunar Surface Imaging	Improve imaging workflow	Testing capture pipelines for high-resolution lunar mosaics	25.75

project_assignee

project_id	person_id
1	1
2	2
3	1

equipment

id	name	type	maintenance_interval
1	Orion XT10	Telescope	180
2	SkyWatcher EQ6-R	Mount	120
3	ZWO ASI2600MC	Camera	90

equipment_booking

id	equipment_id	person_id	start_date	end_date
1	1	1	2025-07-10 20:00	2025-07-10 23:00
2	2	2	2025-07-12 19:30	2025-07-12 22:30
3	3	1	2025-07-15 21:00	2025-07-16 00:00

celestial_target

id	name	type	notes
1	Andromeda Galaxy	Galaxy	Good late-autumn target for wide-field imaging
2	Orion Nebula	Nebula	Bright emission nebula suitable for spectroscopy
3	Jupiter	Planet	Target for planetary imaging during opposition season

observation_session

id	target_id	performed_by	start_date	end_date	notes
1	1	1	2025-10-05 21:00	2025-10-05 23:30	Captured 60 light frames under clear skies
2	2	2	2025-11-02 20:15	2025-11-02 22:45	Collected spectrum readings with calibration frames
3	3	1	2025-12-08 19:00	2025-12-08 20:30	High-frame-rate planetary capture session

observation_data

id	session_id	data_type	file_path	description
1	1	Image	/data/session_1/andromeda_stack.fits	Stacked FITS image for Andromeda session
2	2	Spectrum	/data/session_2/orion_spectrum.csv	Processed emission line spectrum export
3	3	Analysis	/data/session_3/jupiter_quality_report.pdf	Seeing and frame-selection analysis report

maintenance_record

id	equipment_id	maintainer	date	type	notes
1	1	1	2025-06-01	Inspection	Optics cleaned and collimation checked
2	2	2	2025-06-15	Calibration	Mount tracking error recalibrated
3	3	1	2025-06-28	Upgrade	Camera firmware updated and sensor cleaned

equipment_usage_history

id	equipment_id	user	date	purpose
1	1	1	2025-10-05 21:00	Deep-sky imaging of Andromeda Galaxy
2	2	2	2025-11-02 20:15	Stable tracking for Orion spectroscopy
3	3	1	2025-12-08 19:00	Planetary imaging of Jupiter

public_event

id	title	type	date
1	Night Sky Basics	Workshop	2025-08-10 18:00
2	Women in Astronomy Talk	Seminar	2025-09-05 17:30
3	Community Moon Viewing	Public Viewing	2025-10-12 19:30

event_participant

event_id	person_id
1	3
2	2
3	1

partner_institution

id	name
1	North Valley University
2	Cosmic Research Institute
3	City Science Museum

project_collaboration

id	project_id	institution_id	details
1	1	1	Shared access to transit photometry software and student observers
2	2	2	Joint spectral analysis and data review
3	3	3	Public outreach imaging exhibit collaboration

publication

id	title	type	date
1	Transit Signals from a Pilot Exoplanet Survey	Research Paper	2025-05-20
2	Reference Spectra for Bright Nebulae	Report	2025-07-14
3	Optimizing Community Lunar Imaging Sessions	Report	2025-09-01

publication_author

publication_id	author_id
1	1
2	2
3	1